

A R C 3 1 3 3

ARCHITECTURAL VISUALIZATION METHODS 1

COMPUTATIONAL VISUALIZATION + DIGITAL FABRICATION

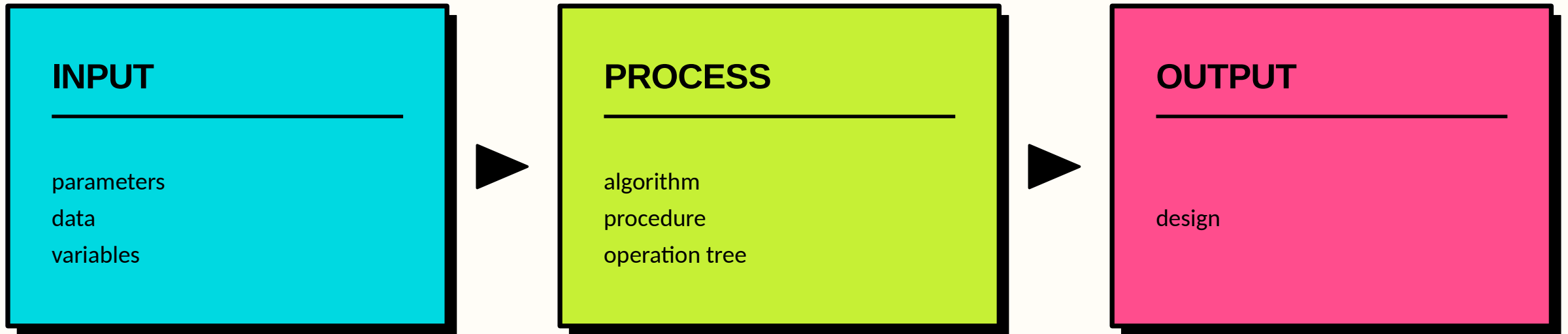
FALL 2026 / TUESDAYS 5:00–7:50 PM / LUIS PACHECO ALCALA

THE DESIGNER IS NO LONGER MAKING CHOICES ABOUT A SINGLE, FINAL OBJECT — BUT CREATING A MATRIX ENCOMPASSING AN ENTIRE POPULATION OF POSSIBLE DESIGNS.

THIS IS WHAT CHANGES WHEN YOU DESIGN WITH CODE.

EVERY COMPUTATIONAL SYSTEM

...has the same three parts. What changes is how much you control the third.



THREE PARADIGMS

The whole course, in one table. Each one extends the last.

01 PROCEDURAL

PROCESS

A fixed sequence
of operations

YOU DEFINE

THE STEPS

OUTPUT IS

**ONE EXPECTED
RESULT**

We are interested primarily in instructions.

MODULE 0

02 PARAMETRIC

PROCESS

Relationships between
inputs and form

YOU DEFINE

THE RULES OF VARIATION

OUTPUT IS

**A FAMILY — YOU
KNOW THE RANGE,
NOT EACH MEMBER**

The same inputs always produce the same output.

MODULES 1–3

03 GENERATIVE

PROCESS

A rule applied to its
own output, repeatedly

YOU DEFINE

THE RULE + THE STARTING STATE

OUTPUT IS

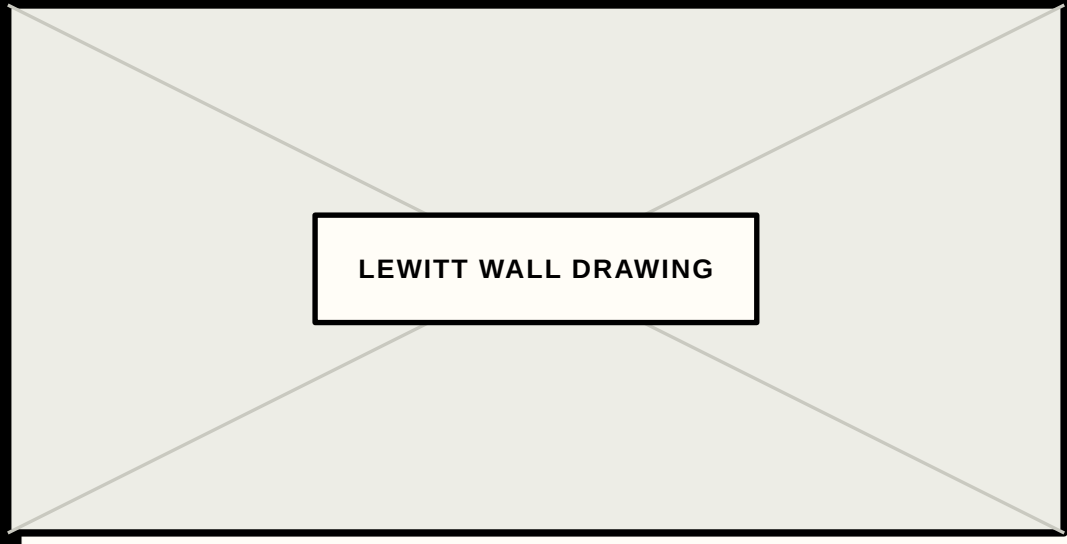
**UNKNOWN, WITHIN
A SPECTRUM YOU SET**

Parameters are iterated by the algorithm itself.

FINAL

01 — PROCEDURAL

Form as a set of instructions. Output: expected.

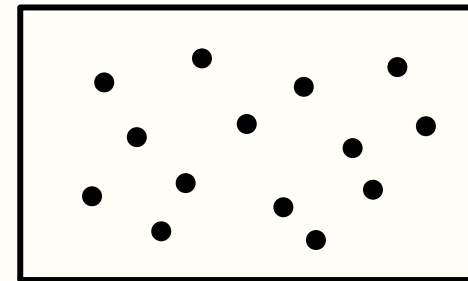


Add: a LeWitt wall drawing beside its written instruction.

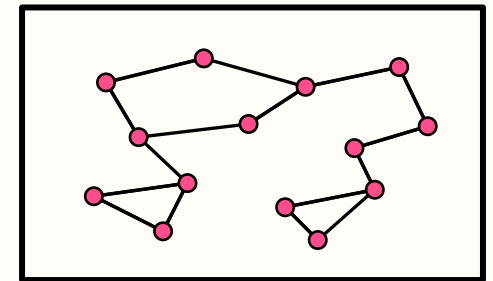
The work IS the instruction. Someone else executes it — LeWitt often never touched the wall.

AN INSTRUCTION

1. Place twenty points at random on the sheet.
2. Connect each point to its two nearest neighbours with a straight line.



1. THE POINTS



2. THE LINES

02 — PARAMETRIC

Parameters drive the form. Output: partially expected.

THE AUTHOR DEFINES THE PROCESS AND ITS FLEXIBILITY.

SAME INPUTS IN



same result out. Always.

CHANGE ONE INPUT

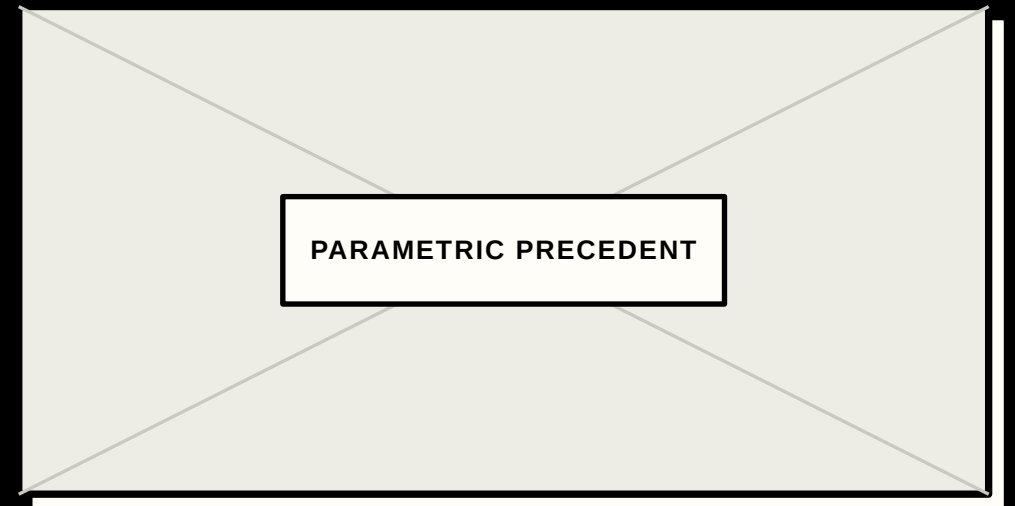


the whole system responds.

NON-DESTRUCTIVE



every step stays editable.

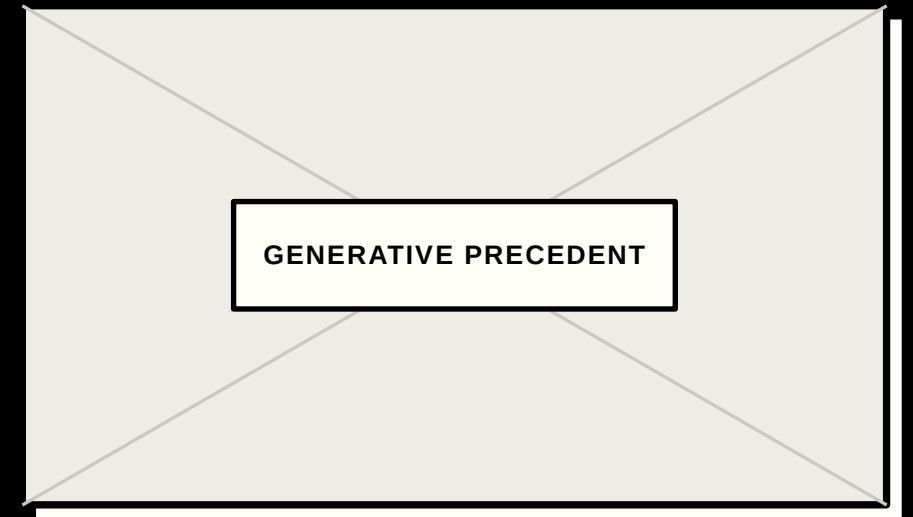
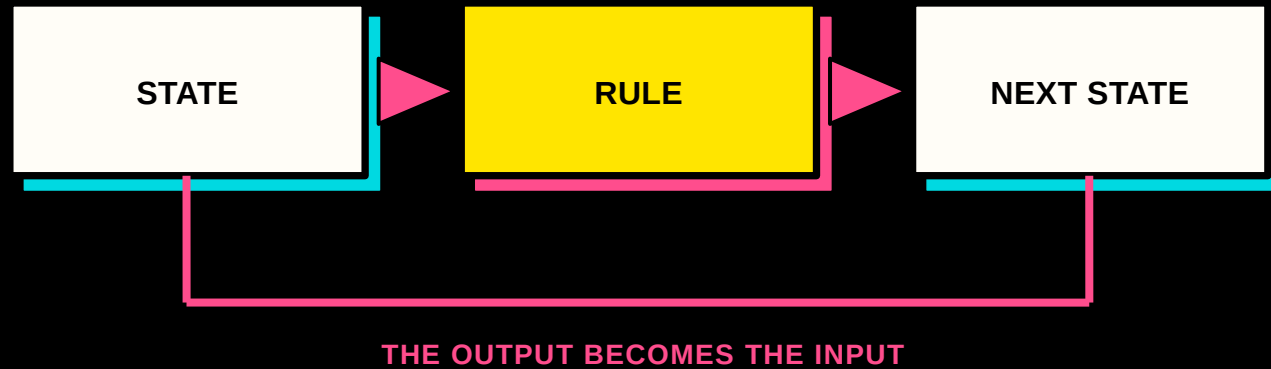


Add: a project whose form is driven by parameters.

03 — GENERATIVE

The system feeds itself. Output: unknown, within a spectrum.

Every generative system is the same three-step loop, run over and over.



Add: an agent trail, a CA, or a growth study.

GROWTH / AGENTS / CELLULAR AUTOMATA / RELAXATION + PACKING

Your final assignment. You will build one of these.

HOW THIS RUNS

Watch before class. Class time is for your problems, not my demo.

01

TUTORIAL ARRIVES

Pre-recorded, sent after the class before. Grasshopper and Blender versions of each.

02

YOU WORK THROUGH IT

Before the session. This is the part that doesn't work if you skip it.

03

WE TROUBLESHOOT

First part of class: what broke. Bring the file, not the description.

04

ASSIGNMENT BRIEFING

Requirements, precedents, then supervised work and critique.

WHAT YOU'LL MAKE

Every module has a drawing and a physical thing.

MODULE 0	PROCEDURE	Massing diagram + written instructions	SMALL 3D PRINT
MODULE 1	FIELDS	Arrays and attractors, systematically	PLOTTER DEMO
MODULE 2	SECTION	Serial sections, contours, hierarchy	LASER-CUT ASSEMBLY
MODULE 3	TESSELLATION	A panelized surface as a filter	3D-PRINTED VAULT
FINAL	GENERATIVE	A tool that iterates + two sheets	—

ISSUED TODAY

0.2 — VISUAL IDENTITY MANUAL

FOUR PAGES. NO MORE. DUE CLASS 4 — SEP 15.

1

Typography — typefaces and hierarchy, with sizes

2

Palette with stated values · line weights with assigned uses

3

The 17" × 11" sheet template, grid shown explicitly

4

A sample sheet, using the system on real work

NO CANVA. NO EXPRESS. NO SLIDES. A template supplies the grid, the hierarchy and the palette — exactly what you're being asked to decide.

BEFORE NEXT CLASS

WATCH

Tutorial 0.1 — Grasshopper / Geometry Nodes basics, transformations, Booleans, non-destructive modeling. Sent tonight.

ENROL

FabLab Safety Orientation on Canvas. Required before you touch any machine — due end of Class 2.

START

Assignment 0.2. Look at how other people build identity systems before you build yours.

BRING

A building you find interesting. Orthogonal massing — no complex curves. You'll rebuild it as a procedure.

PROCEDURAL

EXPECTED

PARAMETRIC

PARTIALLY EXPECTED

GENERATIVE

UNKNOWN, WITHIN A
SPECTRUM YOU DEFINED

QUESTIONS.